Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

AY: 2026-27

Class:	TE-AI&DS	Semester:	V
Course Code:	2015118	Course Name:	Artificial Intelligence & Soft Computing Lab

Name of Student:	Rushikesh Subhash Khemnar
Roll No. :	81
Experiment No.:	7
Title of the Experiment:	Implement Generative AI Techniques Using Open-Source Tools and Frameworks
Date of Performance:	07/09/2026
Date of Submission:	21/09/2026

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission	10	
Total	20	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	4-5	2-3	1
Understanding	4-5	2-3	1
Journal work and timely submission	8-10	5-8	1-4

Checked by

Name of Faculty : Signature :

Date :

Aim To explore and implement Generative AI techniques using open-source tools and frameworks..

Objectives:

After completing this experiment, the student will be able to:

1. To understand the fundamentals of Generative Artificial Intelligence.
2. To explore open-source tools and frameworks used for Generative AI.
3. To implement text generation using a pre-trained Generative AI model.
4. To understand the workflow of prompt-based content generation.
5. To analyze the output generated using different prompts.

Tools Required:

- Hardware:
  - Desktop/Laptop
  - Internet Connection
- Software:
  - Python 3.x
  - Visual Studio Code / Jupyter Notebook
  - Hugging Face Transformers Library
  - Ollama / Google Colab (Optional)
  - Windows/Linux Operating System

Dataset Details:

The experiment utilizes pre-trained Generative AI models available through open-source frameworks. A separate training dataset is not required, as the models are already trained on large-scale text corpora. User-provided prompts are used as input to generate the desired output.

Theory:

Generative Artificial Intelligence (Generative AI) is a branch of AI that enables machines to generate new content such as text, images, code, audio, and videos by learning patterns from large datasets. Modern Generative AI models are based on deep learning architectures, particularly Transformer models, which can understand context and generate coherent outputs from user prompts.

Several open-source frameworks, such as Hugging Face Transformers, Ollama and LangChain, provide pre-trained language models that can be easily integrated into AI applications. These frameworks allow developers to perform tasks such as text generation, summarization, question answering, and code generation without training models from scratch. Generative AI has applications in education, software development, healthcare, content creation and customer support.

Procedure:

1. Install the required Generative AI libraries and frameworks.
2. Configure the development environment.
3. Load a suitable pre-trained language model.
4. Provide input prompts to the model.
5. Execute the program to generate textual responses.
6. Observe the generated outputs for different prompts.
7. Analyze the quality and relevance of the generated content.

Description of the Experiment:

This experiment focuses on exploring open-source Generative AI tools and implementing a pre-trained language model for text generation. A suitable framework is used to load a pretrained model and different prompts are provided to generate meaningful textual responses.

The experiment demonstrates the practical workflow of prompt-based content generation and highlights the capabilities of modern Generative AI models for solving various real-world tasks.

Detailed Description:

Generative AI models available through open-source frameworks enable users to generate meaningful content by providing natural language prompts. In this experiment, a pre-trained language model is accessed through an open-source framework such as Hugging Face or Ollama without performing model training. The model processes the given prompt and generates contextually relevant text based on its learned knowledge.

Different prompts are provided to evaluate the model's ability to generate accurate, coherent and meaningful responses. The generated outputs are observed and analyzed to understand the capabilities and limitations of Generative AI models. This experiment provides practical exposure to prompt-based AI applications and familiarizes students with widely used opensource Generative AI frameworks.

Code:

```
# pyrefly: ignore [missing-import]
from transformers import pipeline, GenerationConfig
print("Loading GPT-2 model...")

generator = pipeline( "text-
generation",
   model="gpt2"
)
print("Model loaded successfully!\n")
generation_config = GenerationConfig(
max_new_tokens=80,
do_sample=True,
   temperature=0.7,
top_p=0.9,
   num_return_sequences=1
)
def generate_text(prompt):

   result = generator(
      prompt,
generation_config=generation_config
   )

   return result[0]["generated_text"]
prompts = [
   "Artificial Intelligence is changing the world because",
   "The future of education will be",
   "Generative AI can be used in healthcare to",
   "Python programming is useful for students because"
]
print("=" * 60)
print("GENERATIVE AI TEXT GENERATION") print("="
* 60)

for i, prompt in enumerate(prompts, 1):

   print(f"\n--- Prompt {i} ---")

   print("Input Prompt:")
   print(prompt)
```

```
    print("\nGenerated          Output:")
output = generate_text(prompt)
    print(output)

    print("-" * 60)
```

Output:

```
Loading GPT-2 model...
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
Loading weights: 100%|                                                  | 148/148 [00:00<00:00, 5411.06it/s]
Model loaded successfully!

============================================================
GENERATIVE AI TEXT GENERATION
============================================================

--- Prompt 1 ---
Input Prompt:
Artificial Intelligence is changing the world because

Generated Output:
[transformers] Ignoring clean_up_tokenization_spaces=True for BPE tokenizer GPT2Tokenizer. The clean_up_tokenization post-processing step is des
igned for WordPiece tokenizers and is destructive for BPE (it strips spaces before punctuation). Set clean_up_tokenization_spaces=False to suppr
ess this warning, or set clean_up_tokenization_spaces_for_bpe_even_though_it_will_corrupt_output=True to force cleanup anyway.
Artificial Intelligence is changing the world because it has become so important to us.

It's amazing. We're seeing that it's also changing the way we think about science.

I want to give you an example.

In a study of a group of computer scientists, the authors found that researchers were more likely to consider artificial intelligence as a threa
t to society than to the traditional scientific community.

They were also
------------------------------------------------------------

--- Prompt 2 ---
Input Prompt:
The future of education will be

Generated Output:
The future of education will be defined by the future of society."

"We are going to build a new school system," she added.

Topics: education, community-and-society, united-kingdom

First posted
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--- Prompt 3 ---
Input Prompt:
Generative AI can be used in healthcare to

Generated Output:
Generative AI can be used in healthcare to better understand patients' health and care needs.

The research has been published in the journal Nature Neuroscience.

"This is a fascinating case of a natural neural network that can be used in medicine to better understand the health of our patients," said co-a
uthor Prof. Jens Stensgaard, from the University of Copenhagen.

"It's important to understand that these models are

--- Prompt 4 ---
Input Prompt:
Python programming is useful for students because

Generated Output:
Python programming is useful for students because it allows you to build software with Python. It's also a great way to learn new languages, and
 it's also a good way to get into the world of software development.

There are a few things you should know about Python, but I won't go into them. I'm going to focus on the most important thing to know: the langu
age.
```

Conclusion:

Generative AI techniques were successfully explored and implemented using open-source tools and frameworks. The experiment demonstrated the practical use of pre-trained language

models for prompt-based text generation and highlighted their applications in solving realworld problems.

Questions to be Answered by Students:

1. Which open-source framework was used to implement the Generative AI model? Answer: Hugging Face Transformers framework was used to implement the Generative AI model.

2. Which pre-trained model was used for text generation? Answer: The GPT-2 (Generative Pre-trained Transformer 2) model was used for text generation.

3. What type of output was generated for the given input prompts?

Answer: The model generated textual content/completions based on the given naturallanguage input prompts.

4. State two observations obtained from the implementation of the Generative AI model. Answer:
   1. Different input prompts produced different generated text, showing that the output depends on the context of the prompt.
   2. The generated text was generally coherent and contextually related, but some outputs were incomplete or less relevant, showing the limitations of the GPT-2 model.